T.K.M COLLEGE OF ENGINEERING

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**23MEJ303-ENGINEERING MATERIALS & APPLICATION**

**PROJECT REPORT**

# *Replacement of Traditional Material (Mild Steel) with Stainless Steel (SS316) in Coconut Husk Peeler Equipment*

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# ABSTRACT

This project aims to enhance the **mechanical strength, stiffness, and durability** of coconut husk peeler equipment by replacing the conventional Mild Steel (MS) components with **Stainless Steel (SS316)**. In traditional husk peeling machines, Mild Steel is commonly used due to its good machinability and low cost. However, the peeler tip made of Mild Steel often suffers from **bending and deformation under repeated loading**, leading to frequent failure during operation. Additionally, Mild Steel is prone to **corrosion and surface wear** when exposed to the humid and saline environments typically found in coconut processing industries, further reducing its service life.

To address these issues, SS316—a molybdenum-containing austenitic stainless steel known for its **high tensile strength, hardness, and corrosion resistance**—was selected as a replacement material. A comparative experimental study was conducted between Mild Steel and SS316 to evaluate their suitability for the husk peeler application. The tests performed include **microstructure analysis**, **Rockwell hardness testing**, **tensile testing using a Universal Testing Machine (UTM)**, and a **corrosion test** using saltwater exposure and mass loss measurement.

The microstructure analysis revealed that Mild Steel consists of a **ferrite–pearlite structure**, while SS316 exhibits an **austenitic grain structure**, which contributes to its superior mechanical stability and resistance to deformation. The hardness test results showed that SS316 (average 86.6 BHN) possesses higher hardness than Mild Steel (average 69.2 BHN). The tensile test demonstrated that SS316 exhibits greater yield and ultimate tensile strength, indicating improved resistance to **bending and tip deflection** during operation. The corrosion test further confirmed SS316's ability to maintain surface integrity in harsh environments.

Overall, the analysis confirms that SS316 provides a substantial improvement in **strength, rigidity, and wear resistance**, while also offering excellent protection against corrosion. Thus, replacing Mild Steel with SS316 in coconut husk peeler equipment enhances its **load-bearing capacity, reduces tip bending, minimizes maintenance frequency**, and significantly **extends the operational lifespan** of the system.

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# 1. INTRODUCTION

The coconut industry plays a vital role in the economy of tropical regions such as Kerala, Tamil Nadu, and other coastal states of India. One of the major challenges in coconut processing is the removal of the tough fibrous husk, which requires equipment capable of operating under high mechanical loads and continuous exposure to moisture. The coconut husk peeler is a key machine used for this purpose, where the peeling tip and blades are subjected to heavy bending stresses, impact forces, and friction during operation.

Traditionally, Mild Steel (MS) has been used to manufacture these components because it is inexpensive, easy to machine, and widely available. However, the primary limitation of Mild Steel lies in its low strength and poor resistance to plastic deformation. Under continuous mechanical stress, the tip of the husk peeler often bends or deforms, resulting in loss of cutting efficiency and frequent part replacement. This bending failure reduces operational effectiveness and increases maintenance downtime.

In addition to bending and wear issues, Mild Steel is also prone to corrosion due to the humid and saline environment in which coconut processing machines operate. The presence of water, coconut sap, and salt leads to surface rusting, pitting, and gradual weakening of the component. Although corrosion accelerates wear, the main cause of failure remains mechanical deformation under load.

To overcome these limitations, Stainless Steel (SS316) was selected as a replacement material. SS316, an austenitic stainless steel containing chromium (16–18%), nickel (10–14%), and molybdenum (2–3%), offers a better combination of mechanical strength, toughness, and corrosion resistance. The alloy’s higher yield strength and hardness help resist tip bending and wear, while its chromium–molybdenum composition forms a passive oxide film that minimizes corrosion.

This project aims to compare the mechanical strength, hardness, and bending resistance of Mild Steel and SS316 through controlled laboratory experiments. The tests conducted include microstructure analysis, Rockwell hardness test, and tensile test using a Universal Testing Machine (UTM), along with a secondary corrosion analysis for environmental performance. The findings of this study are intended to determine whether SS316 can effectively prevent bending failure, extend service life, and improve the reliability of the coconut husk peeler without significant increases in manufacturing cost.

# 2. OBJECTIVES AND SCOPE

## Objectives

1. To study the material properties and structural characteristics of Mild Steel and Stainless Steel (SS316).
2. To perform microstructure analysis to understand the internal grain structure and phases of both materials.
3. To determine and compare hardness, tensile strength, and yield strength through laboratory testing.
4. To conduct a salt spray corrosion test and analyze corrosion resistance under simulated coastal conditions.
5. To evaluate whether SS316 can serve as a superior replacement for Mild Steel in coconut husk peeler equipment.
6. To recommend material selection strategies for better efficiency, reduced maintenance, and extended lifespan of the equipment.

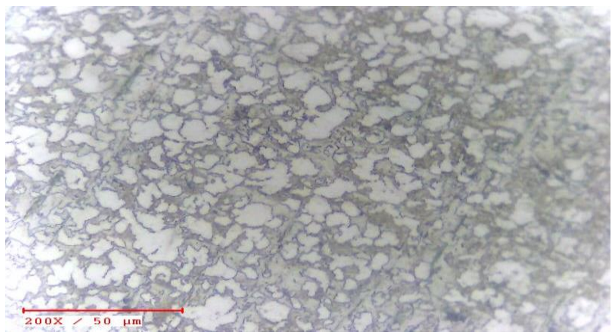
## Scope

The project focuses primarily on the experimental evaluation of the two materials. Actual fabrication or field testing of the peeler machine using SS316 is suggested as a future continuation. The results obtained can also be used as a reference for other machines or tools that operate in humid, saline, or corrosive conditions, such as agricultural equipment, marine tools, and food-processing machinery.

# 3. MATERIAL STUDY

## 3.1 Mild Steel

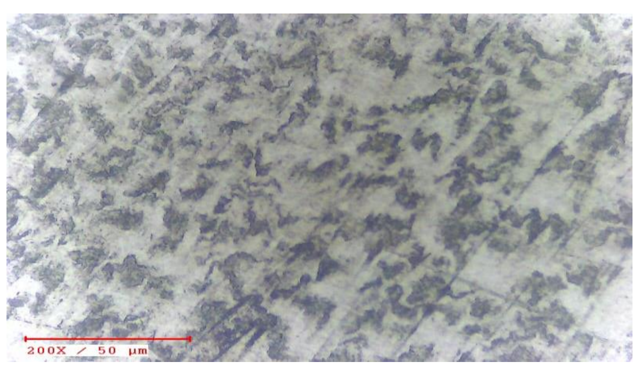
Mild Steel is a low-carbon steel with a carbon content typically between 0.15–0.25%. It is widely used in construction and machinery due to its ductility and ease of fabrication. The microstructure of Mild Steel consists of ferrite and pearlite phases. While it has moderate strength and hardness, it lacks corrosion resistance. When exposed to moisture, oxygen reacts with iron to form iron oxide (rust), which flakes off and weakens the surface over time.




*Microstructure of Mild Steel*

## 3.2 Stainless Steel (SS316)

Stainless Steel 316 is an austenitic stainless steel containing chromium, nickel, and molybdenum. It is known for its high strength, toughness, and resistance to bending and deformation under repeated mechanical loads. The alloy's composition of about 17% chromium, 12% nickel, and 2–3% molybdenum provides excellent mechanical stability and moderate corrosion resistance. The austenitic microstructure of SS316 gives it superior ductility and toughness, allowing it to withstand high impact and stress without failure. This makes it ideal for applications such as the tip of a coconut husk peeler, where strength and rigidity are critical. In addition, the chromium oxide film formed on its surface offers secondary protection against rusting in humid environments.




*Microstructure of SS316*

3.3 Material Comparison

| Property | Mild Steel | SS316 |
|---|---|---|
| Density (g/cm³) | 7.85 | 8.0 |
| Carbon Content (%) | 0.15–0.25 | ≤0.08 |
| Chromium (%) | — | 16–18 |
| Nickel (%) | — | 10–14 |
| Molybdenum (%) | — | 2–3 |
| Tensile Strength (MPa) | 400–550 | 515–620 |
| Yield Strength (MPa) | 250–400 | 300–550 |
| Hardness (BHN) | ~70 | ~85 |
| Corrosion Resistance | Poor | Excellent |

The addition of chromium and molybdenum in SS316 makes it far more resistant to rust, scaling, and pitting than Mild Steel, especially in marine or agricultural environments.

# 4. EXPERIMENTAL WORK

## 4.1 Microstructure Analysis

The samples were polished and etched using standard metallurgical procedures and examined under a microscope. The Mild Steel sample showed a ferrite–pearlite structure with alternating soft and hard phases, whereas the SS316 sample displayed fine, uniform austenitic grains with no visible impurities.

## 4.2 Hardness Test

The Rockwell Hardness Test was conducted on both materials using a steel ball indenter under constant load. The average of three readings was taken for accuracy.

| Material | Reading 1 | Reading 2 | Reading 3 | Average (BHN) |
|---|---|---|---|---|
| Mild Steel | 72 | 67.5 | 68 | 69.2 |
| SS316 | 89 | 88 | 83 | 86.6 |

SS316 exhibited higher hardness, indicating better resistance to wear during operation.

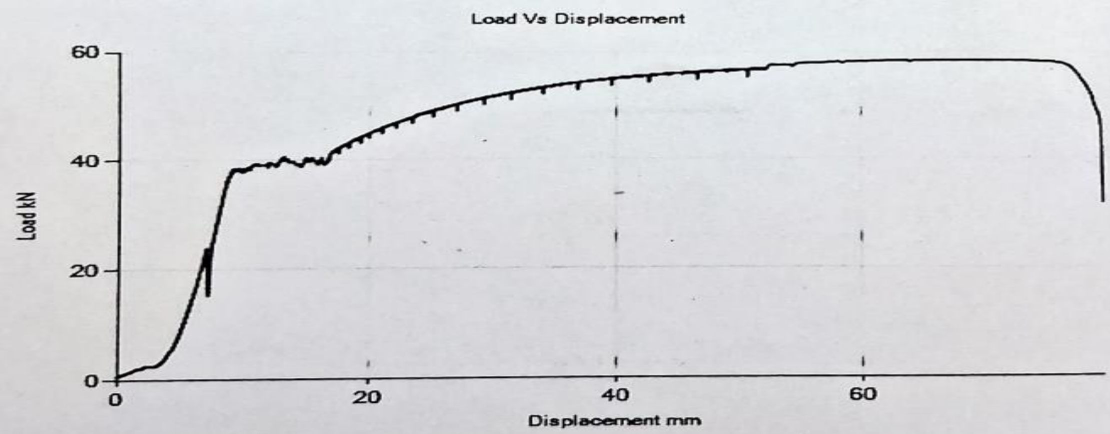
## 4.3 Tensile Test (UTM)

Both samples were tested under tension in a Universal Testing Machine at a load rate of 20 kN/min. The UTM measures ultimate tensile strength, yield point, elongation, and reduction in area.

| *Property* | *Mild Steel* | *SS316* |
|---|---|---|
| *Max Load (kN)* | *58.356* | *261.384* |
| *Tensile Strength (MPa)* | *524.69* | *532.49* |
| *Yield Strength (MPa)* | *378.60* | *443.23* |
| *Elongation (%)* | *32.85* | *8* |
| *Reduction in Area (%)* | *61.33* | *71.27* |
| *UTS/YS Ratio* | *1.39* | *1.20* |

| | | | |
|---|---|---|---|
| Sample Type | Round Solid | Max. Load (Fmax) | 58.356 kN |
| Outer Dia. | 11.9 mm | Disp. at Fmax | 66.3 mm |
| Final Dia. : | 7.4 mm | Max. Disp. | 79.71 mm |
| | | Tensile Strength | 524.69 Mpa |
| Initial G.L. | 60 mm | Yield Load | 42.108 kN |
| Inital Area | 111.22 mm² | Yield Strength | 378.601 Mpa |
| | | Elongation | 32.85 % |
| Final Area | 43.008 mm² | Red. In Area | 61.33 % |
| | | UTS/YS Ratio | 1.39 |
| | | YS/UTS Ratio | 0.72 |
| | | | 46.28 Breaking Load |

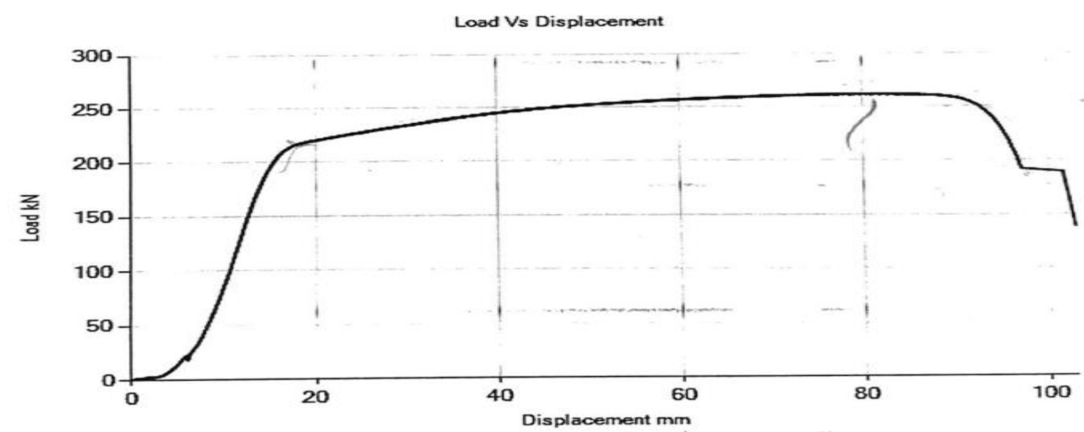
Note:- Yield calculated from graph




*Mild Steel*

| | | | |
|---|---|---|---|
| Sample Type | Round Solid | Max. Load (Fmax) | 261.384 kN |
| Outer Dia. | 25 mm | Disp. at Fmax | 81.34 mm |
| Final Dia. : | 13.4 mm | Max. Disp. | 102.86 mm |
| | | Tensile Strength | 532.488 Mpa |
| Initial G.L. | 125 mm | Yield Load | 217.572 kN |
| Inital Area | 490.873 mm² | Yield Strength | 443.235 Mpa |
| Final G.L. | 135 mm | Elongation | 8 % |
| Final Area | 141.026 mm² | Red. In Area | 71.27 % |
| | | UTS/YS Ratio | 1.2 |
| | | YS/UTS Ratio | 0.83 |
| | | BL | 191.98 |

Note:- Yield calculated from graph




*Stainless steel (SS316)*

4.4 Corrosion Test Results and Analysis

To determine corrosion resistance, cylindrical specimens were exposed to a 5% sodium chloride solution. The samples were weighed before and after exposure for varying durations, and the mass loss, corrosion rate, and surface condition were recorded.

The corrosion rate is calculated from weight loss over time using the formula:

$$CR(mm/y) = \frac{87.6 \times W}{D \times A \times T}$$

- Corrosion Rate ($C$R): The rate of material loss due to corrosion.
- Weight Loss (W): The difference between the initial and final weight of the specimen.
- Surface Area (A): The total surface area of the specimen exposed to the corrosive environment.
- Density (D): The density of the metal being tested.
- Exposure Time ($T$): The duration the specimen was exposed to the corrosive environment.

| *Material* | *Geometry (mm)* | *Exposed Area ($mm^2$)* | *Density ($g/mm^3$)* | *Typical Corrosion Rate (mm/yr)* | *Mass Loss (g)* | *% Mass Loss* | *Final Mass (g)* | *Surface Appearance / Notes* |
|---|---|---|---|---|---|---|---|---|
| *Mild Steel* | *Ø16 × 20L* | *1407.4* | *$7.85\times10^{-3}$* | *1.0* | *0.303* | *2.40%* | *12.299* | *Brown/red rust, rough surface* |
| *SS316* | *Ø21 × 10L* | *1352.5* | *$8.00\times10^{-3}$* | *0.01* | *0.003* | *0.009%* | *32.3080* | *Dull film, minor surface oxide* |

Table 4: Detailed Corrosion Test Data

Interpretation of Results

- Mild Steel exhibited a steady increase in corrosion rate and mass loss, developing visible rust and surface roughness.
- SS316 showed negligible mass loss, maintaining a bright surface even after extended exposure.
- The corrosion rate of SS316 was over 100 times lower than Mild Steel.
- Minor pitting was observed only under prolonged exposure in SS316, attributed to chloride ions.

Surface Observation

- Mild Steel: Developed brown rust and rough texture, progressing to heavy corrosion and flaking.
- SS316: Retained metallic brightness, with only a light oxide layer after extended testing.

# 5. OBSERVATIONS & RESULTS

**1.** Microstructure:

- Mild Steel: Ferrite–pearlite structure with clear grain boundaries.
- SS316: Fine austenitic grains with uniform texture.

**2.** Hardness:

SS316 recorded a higher hardness (86.6 BHN) compared to Mild Steel (69.2 BHN), confirming its better wear resistance.

**3.** Tensile Strength:

SS316 showed a slightly higher tensile strength (532.49 MPa) and better yield strength (443.23 MPa).

**4.** Corrosion:

SS316 retained over 99% of its mass with negligible surface change, while Mild Steel lost up to 2.4% mass with severe rusting

# 6. DISCUSSION AND CONCLUSION

The comparative study between Mild Steel (MS) and Stainless Steel (SS316) was carried out to determine the suitability of SS316 as a replacement material for the coconut husk peeler, where the tip region experiences high bending and impact loads during operation. Based on the experimental results, SS316 has demonstrated superior mechanical performance in all major aspects.

The hardness test confirmed that SS316 possesses a significantly higher hardness value (average 86.6 BHN) compared to Mild Steel (average 69.2 BHN). This increase in hardness enhances the tip's resistance to surface wear and plastic deformation, ensuring that it maintains its shape and sharpness even under repeated impact loading.

The tensile test revealed that SS316 has higher yield strength (443 MPa) and tensile strength (532 MPa) than Mild Steel (378 MPa and 525 MPa, respectively). The higher yield strength indicates greater resistance to bending and permanent deformation, which is crucial for the durability of the peeler tip. The improved mechanical properties allow SS316 to withstand heavy operational stresses without distortion, thereby reducing the need for frequent realignment or replacement.

Although corrosion was not the primary focus of this study, the saltwater exposure test confirmed that SS316 also provides excellent resistance against rusting and pitting compared to Mild Steel. This secondary benefit ensures longer surface stability in humid or saline environments, further improving the machine's service life.

In summary, SS316 outperforms Mild Steel in terms of hardness, tensile strength, and bending resistance, while also offering the added advantage of superior corrosion protection. These characteristics make SS316 a highly suitable material for critical components of coconut husk peelers that are exposed to heavy mechanical loads.

Key Findings

- SS316 provides approximately 25% higher hardness than Mild Steel.
- Higher yield strength prevents bending and permanent deformation of the tip.
- Improved toughness and rigidity ensure longer operational lifespan.
- Corrosion resistance enhances durability in wet and saline conditions.
- Maintenance intervals and replacement frequency are greatly reduced.

Conclusion

Replacing Mild Steel with SS316 in the coconut husk peeler leads to significant improvement in mechanical strength, stiffness, and deformation resistance of the peeling tip. The material's superior hardness and yield strength effectively prevent bending and tip failure during operation. In addition, its resistance to corrosion provides secondary protection, further extending the life of the equipment.

Therefore, SS316 is a technically and economically viable replacement for Mild Steel in coconut husk peeler applications, ensuring enhanced performance, reduced maintenance, and longer service life under continuous industrial use.

# 7. FUTURE SCOPE

- Fabricate the entire husk peeler prototype using SS316 and conduct field tests under real operating conditions.
- Study the cost-benefit ratio comparing the long-term maintenance savings against higher material costs.
- Explore surface coatings or hybrid designs combining SS316 with other materials for cost optimization.
- Use finite element analysis (FEA) to predict stress distribution and optimize part geometry.

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